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Rahul Bhaumik, PhD

Designer and HCI researcher — soft, tangible interfaces

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PROFILE

Designer and HCI researcher exploring how form, surface and material shape the way we touch, press and interact with technology. Trained as an architect and product designer, I design and fabricate soft, passive interfaces, from embroidered textiles to 3D-printed structures that sense pressure and deformation, and study how their affordances guide interaction.

Research interests: human-computer interaction · soft and tangible interfaces · smart textiles · 3D-printed sensing · automotive interfaces · extended reality · design pedagogy · AI in design.

APPOINTMENTS

2026 – present
Researcher
Media Interaction Lab, Faculty of Engineering, Free University of Bozen-Bolzano (unibz), Italy

May – Oct 2025
Visiting Researcher
Human-Computer Interaction Institute (HCII), Carnegie Mellon University, Pittsburgh, USA

2023 – 2026
Doctoral Researcher, PhD in Computer Science (38th cycle)
Free University of Bozen-Bolzano, Italy · design, testing, analysis and dissemination of research on tangible, soft interfaces. Funded by the European Union NextGenerationEU (PNRR, Mission 4, Component 2, Investment 3.3).

2018 – 2023
Assistant Professor
Faculty of Architecture & Design, PES University, Bangalore, India

- Taught product and interaction design courses in the Bachelor of Design (B.Des) programme
- Coordinator and guide for the B.Des thesis / capstone projects
- Research coordinator of the department; multidisciplinary research in design, XR, assistive technology and pedagogy

EDUCATION

2023 – 2026
PhD in Computer Science
Free University of Bozen-Bolzano, Italy · defended 12 June 2026

Thesis: *Affordances of Soft Interfaces: A Design and Fabrication Approach*. Supervisors: Prof. Michael Haller, Prof. Antonella De Angeli.

2016 – 2018
Master of Design (M.Des), Product Design & Engineering
Indian Institute of Science (IISc), Bangalore, India · MHRD scholarship (GATE)

2010 – 2015
Bachelor of Architecture (B.Arch)
School of Planning and Architecture, New Delhi, India

TEACHING

Assistant Professor, B.Des programme, PES University, Bangalore (2018 – 2023). Courses taught:

Capstone	**Program End Design Project** Undergraduate capstone across product, interaction and communication design · 70+ students per year · programme coordinator and thesis guide
~30 / sem.	**Game Design** Fundamentals of game design; MDA, AGE and DPE frameworks; serious games and gamification; prototyping board and app-based games
~50 / sem.	**Ergonomics** Anthropometry, posture, movement and grip; RULA/REBA; cognitive ergonomics: attention, memory and perception; visual ergonomics
~30 / sem.	**Interaction Design for Mixed Reality** Unity 3D modelling and scripting; VR headsets; design thinking strategies for mixed-reality applications
~30 / sem.	**Basic Computing in Design** Generative design with Rhino and Grasshopper; digital fabrication
~30 / sem.	**Product / Interaction Design Project Studio** User research, need analysis, ideation, mock-ups and functional prototyping, user testing
Curriculum	■ Developed course structures, assignments and lab work for multiple design courses ■ Revised, restructured and introduced courses following AICTE and NEP (India) guidelines

PUBLICATIONS

JOURNAL ARTICLES

01 A. Ahmed, **R. Bhaumik**, R. Didero, M. Haller. **Soft, Scalable Spacer-Knitted Piezoresistive Pressure Sensors with Tunable Localized Sensitivity Enabled by Heat-Expandable Yarn Inlays.** IEEE Sensors Letters 10(7), 2026. doi:10.1109/LSENS.2026.3699462

02 **R. Bhaumik**, T. Preindl, A. Ion, C. Ayala-Garcia, N. Cohen, M. Haller, N. Münzenrieder. **Inductive Pressure Sensors Using 3D-Printed Structures With Tunable Stiffness.** IEEE Sensors Letters 9(5), 2025. doi:10.1109/LSENS.2025.3562455

03 P. Lugoda, E. S. Oliveros-Mata, K. Marasinghe, **R. Bhaumik**, N. Pretto, C. Oliveira, T. Dias, T. Hughes-Riley, M. Haller, N. Münzenrieder, D. Makarov. **Submersible touchless interactivity in conformable textiles enabled by highly selective overbraided magnetoresistive sensors.** Communications Engineering 4(1), 2025. doi:10.1038/s44172-025-00373-x

04 K. F. Mwaita, **R. Bhaumik**, A. Ahmed, A. Sharma, A. De Angeli, M. Haller. **Emotion-Aware In-Car Feedback: A Comparative Study.** Multimodal Technologies and Interaction 8(7), 54, 2024. doi:10.3390/mti8070054

05 U. Sait, G. Lal K.V., S. Shivakumar, T. Kumar, **R. Bhaumik**, S. Prajapati, K. Bhalla, A. Chakrapani. **A deep-learning based multimodal system for Covid-19 diagnosis using breathing sounds and chest X-ray images.** Applied Soft Computing 109, 107522, 2021. doi:10.1016/j.asoc.2021.107522

CONFERENCE PAPERS AND BOOK CHAPTERS

06 **R. Bhaumik**, C. Ayala-Garcia, N. Münzenrieder, M. Haller, A. Ion. **InSense3D: Designing Smart 3D-Printed Structures Leveraging Ferromagnetic Filaments for Inductive Deformation Sensing.** Proc. ACM CHI Conference on Human Factors in Computing Systems (CHI '26), 2026. doi:10.1145/3772318.3791824

07 H. K. Gupta, V. Gadgihalli, **R. Bhaumik**, S. Shirsi, D. Gupta, R. H. Meena, S. Varun. **Gamified Ideation Board (GIB) for Creative Problem-Solving.** ICoRD 2025, Lecture Notes in Mechanical Engineering, Springer, pp. 183–194, 2025. doi:10.1007/978-981-96-5495-6_14

08 **R. Bhaumik**, K. F. Mwaita, F. Solaroli, A. Ahmed, A. De Angeli, M. Haller. **Interaction with a 3D Surface for an Innovative Input Experience on a Central Console.** Proc. AutomotiveUI '24, ACM, pp. 136–148, 2024. doi:10.1145/3640792.3675707

09 A. Seetharam, T. Kumar, H. Gupta, **R. Bhaumik**. **A Marker-Based Augmented Reality Application to Teach Chemistry to Indian Students.** SmartCom 2024, LNNS 947, Springer, pp. 415–435, 2024. doi:10.1007/978-981-97-1326-4_34

10 **R. Bhaumik**, S. Prajapati, T. Kumar, K. Bhalla, S. S. Ashok. **Smart Vernacular Architecture: A Framework for Assessment and Virtual Reality-based Visualisation of Indigenous Toda Dwellings.** Procedia Computer Science 218, pp. 651–670, 2023. doi:10.1016/j.procs.2023.01.047

11 **R. Bhaumik**, T. Kumar, U. Sait. **An Intelligent Virtual Environment for Designers with Reduced Motor Abilities.** Procedia Computer Science 218, pp. 488–503, 2023. doi:10.1016/j.procs.2023.01.031

12 S. P. Prajapati, **R. Bhaumik**, T. Kumar. **An Intelligent ABM-based Framework for Developing Pandemic-Resilient Urban Spaces in Post-COVID Smart Cities.** Procedia Computer Science 218, pp. 2299–2308, 2023. doi:10.1016/j.procs.2023.01.205

13 S. Prasad, **R. Bhaumik**, S. Jamadagni, M. Narasimha. **Addressing Online-Learning Challenges Through Smartphone-Based Gamified Learning Platform.** ICL 2022, LNNS 633, Springer, pp. 990–1001, 2023. doi:10.1007/978-3-031-26876-2_93

14 D. Gopalakrishnan, P. Warale, **R. Bhaumik**. **Design of a Maternal Healthcare Monitoring Device for Pregnant Women.** Design for Tomorrow, Vol. 3, Smart Innovation, Systems and Technologies 223, Springer, pp. 495–509, 2021. doi:10.1007/978-981-16-0084-5_40

15 S. P. Prajapati, **R. Bhaumik**, T. Kumar, U. Sait. **An AI-Based Pedagogical Tool for Creating Sketched Representation of Emotive Product Forms in the Conceptual Design Stages.** ICT Systems and Sustainability, Springer, pp. 649–659, 2020. doi:10.1007/978-981-15-8289-9_63

16 A. Chakrapani, T. Kumar, S. Shivakumar, **R. Bhaumik**, K. Bhalla, S. Prajapati. **A Pandemic-specific "Emergency Essentials Kit" for Children in the Migrant BoP Communities.** IEEE International Smart Cities Conference (ISC2), 2020. doi:10.1109/ISC251055.2020.9239078

17 U. Sait, V. Ravishankar, T. Kumar, **R. Bhaumik**, G. L. K.V., K. Bhalla, K. S. Sanjay. **Design and development of an assistive device for the visually impaired.** Procedia Computer Science 167, pp. 2244–2252, 2020. doi:10.1016/j.procs.2020.03.277

18 G. L. K.V., U. Sait, T. Kumar, **R. Bhaumik**, S. Shivakumar, K. Bhalla. **Design and development of a smartphone-based application to save lives during accidents and emergencies.** Procedia Computer Science 167, pp. 2267–2275, 2020. doi:10.1016/j.procs.2020.03.279

19 **R. Bhaumik**, S. P. Prajapati, T. Kumar, V. Mishra, K. Bhalla. **An Inclusive Community Based Water Purification and Monitoring System for the Base of the Pyramid.** IEEE Global Humanitarian Technology Conference (GHTC), 2019. doi:10.1109/GHTC46095.2019.9033102

20 U. Sait, V. D. Ravishankar, T. Kumar, S. Shivakumar, G. L. K.V., K. Bhalla, M. Singh, **R. Bhaumik**. **Memhans: an Assistive Device for Dementia Patients.** IEEE Global Humanitarian Technology Conference (GHTC), 2019. doi:10.1109/GHTC46095.2019.9033140

21 A. N. Bhatt, **R. Bhaumik**, K. Moothedath Chandran, A. Chakrabarti. **IISc DBox: A Game for Children to Learn Design Thinking.** ASME IDETC-CIE, Vol. 59216, 2019. doi:10.1115/DETC2019-98066

22 U. Sait, G. L. K.V., T. Kumar, **R. Bhaumik**, K. Bhalla. **A framework outlining a daylight responsive model for smart buildings.** Journal of Physics: Conference Series 1343(1), 012166, 2019. doi:10.1088/1742-6596/1343/1/012166

23 **R. Bhaumik**, A. Bhatt, M. C. Kumari, S. R. Menon, A. Chakrabarti. **A Gamified Model of Design Thinking for Fostering Learning in Children.** ICoRD '19, Research into Design for a Connected World, Springer, pp. 1023–1036, 2019. doi:10.1007/978-981-13-5977-4_85

24 A. Agarwal, D. V. Jeevitha Shree, K. Saluja, A. Sahay, P. Mounika, A. Sahu, **R. Bhaumik**, V. Kumar Rajendran, P. Biswas. **Comparing Two Webcam-Based Eye Gaze Trackers for Users with Severe Speech and Motor Impairment.** ICoRD '19, Research into Design for a Connected World, Springer, pp. 641–652, 2019. doi:10.1007/978-981-13-5977-4_54

PATENT

2023 **System and Method for Identifying Problems and Corresponding Solutions While Performing an Activity**
Co-inventor · Indian Patent No. 455667, granted 29 September 2023

AWARDS AND FUNDING

2023 – 2026 **Doctoral scholarship, European Union NextGenerationEU (PNRR)**
Free University of Bozen-Bolzano

2019 **IEEE Humanitarian Activities Committee (HAC) funding**
To present humanitarian technology research at IEEE GHTC 2019, Seattle, USA

2019 **Distinguished Paper Award, ICoRD 2019**
A Gamified Model of Design Thinking for Fostering Learning in Children

2016 – 2018 **MHRD scholarship, Government of India**
Fully funded Master's at IISc Bangalore, awarded on GATE qualification

SELECTED PRESENTATIONS

2026 **ACM CHI 2026, Barcelona, Spain**
InSense3D: Designing Smart 3D-Printed Structures Leveraging Ferromagnetic Filaments for Inductive Deformation Sensing

2025 **IEEE FLEPS 2025, NUS, Singapore**
Inductive Pressure Sensors Using 3D-Printed Structures With Tunable Stiffness

2024 **ACM AutomotiveUI 2024, Stanford University, USA**
Interaction with a 3D Surface for an Innovative Input Experience on a Central Console

2022 **ICMLDE 2022 (online)**
Three papers: Intelligent Virtual Environment for Designers; Smart Vernacular Architecture; ABM-based Framework for Pandemic-Resilient Urban Spaces

2019	**IEEE GHTC 2019, Seattle, USA** An Inclusive Community Based Water Purification and Monitoring System for the Base of the Pyramid
2019	**ICoRD 2019, Bengaluru, India** A Gamified Model of Design Thinking for Fostering Learning in Children

SKILLS

FABRICATION	3D printing with flexible and ferromagnetic filaments; lattice and deformable structures; embroidery and sewing of functional textiles; sensor housings and functional prototypes
SENSING	Inductive and piezoresistive sensing; capacitive sensing with conductive embroidery; sensor characterisation and empirical validation; Arduino
DESIGN & CODE	Rhino and Grasshopper (parametric, fabrication-oriented modelling); Python (generative design, data analysis, visualisation); Unity and C#; Figma; AutoCAD; Revit
METHODS	User studies and gesture elicitation; design thinking; usability evaluation; quantitative and qualitative analysis

MEMBERSHIPS

ACM · IEEE	Student member
COA INDIA	Registered Architect, Council of Architecture (since 2016)

LANGUAGES

ENGLISH	Proficient (C2)
BENGALI	Native
HINDI	Working proficiency